Tensor-Algebra Example: Finite-Dimensional Obstruction with Universal Injectivity

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Abstract

We construct an explicit R-bimodule M (for $R = k[\alpha,\beta]/(\alpha,\beta)^2$) which admits no finite-dimensional realization inside $\mathrm{End}_k(N)$ compatible with the distinguished element $e \in M$, yet whose canonical map into the universal quotient $T_R(M)/(e-1)$ is injective.

Setup. Let

$$R := k[\alpha,\beta]/(\alpha,\beta)^2, \qquad \mathfrak{m} := (\alpha,\beta), \qquad \mathfrak{m}^2 = 0.$$

Let M be the k-vector space with basis

$$\{e, u, v, x, y, z, p, q, w\}.$$

Embed $R \hookrightarrow M$ by $1 \mapsto e,\ \alpha \mapsto u,\ \beta \mapsto v$, and let $1 \in R$ act as identity on both sides. Define the $\mathfrak{m}$-actions on basis elements by the following *only* nonzero ones:

$$\begin{aligned} &\text{(left)} && \alpha e = u,\ \beta e = v,\ \alpha x = v,\ \beta x = q, \\ &\text{(right)} && e\alpha = u,\ e\beta = v,\ y\beta = u,\ y\alpha = w,\ z\alpha = p, \end{aligned}$$

and declare every other action of α, β on basis vectors to be 0 (in particular $z\beta = 0$, $x\alpha = 0$, $\alpha y = 0$, etc.).

Lemma 1. *M is an R-bimodule and $R \hookrightarrow M$ is a bimodule monomorphism.*

Proof. Since 1 acts as identity and $\mathfrak{m}^2 = 0$, it suffices to check the associativity relations

$$(r_1 r_2)m = r_1(r_2 m), \qquad m(r_1 r_2) = (m r_1) r_2, \qquad (r_1 m) r_2 = r_1 (m r_2)$$

for $r_1, r_2 \in \{\alpha, \beta\}$.

All nonzero left actions land in $\{u, v, q\}$, on which the left and right α,β-actions are 0; thus $r_1(r_2 m) = 0$ and $(r_1 m) r_2 = 0$.

All nonzero right actions land in $\{u, v, w, p\}$, on which the left and right α,β-actions are 0; thus $(m r_1) r_2 = 0$ and $r_1(m r_2) = 0$.

Therefore all required associativity conditions hold. The embedding respects actions by construction and is injective since $\{e, u, v\}$ are linearly independent. $\square$

Let $T_R(M) := \bigoplus_{n \geq 0} M^{\otimes_R n}$ be the tensor algebra ($M^{\otimes 0} = R$). Write $\iota : M \to T_R(M)/(e-1)$ for the degree-1 inclusion followed by the quotient.

Finite-dimensional obstruction

Proposition 1 (No finite-dimensional realization)**.** *There is no finite-dimensional left R-module N and injective R-bimodule map $f : M \hookrightarrow \mathrm{End}_k(N)$ with $f(e) = \mathrm{id}_N$, where $\mathrm{End}_k(N)$ is an R-bimodule via*

$$r \cdot T \cdot s := \phi(r)\, T\, \phi(s)$$

for the left action $\phi : R \to \mathrm{End}_k(N)$.

Proof. Set $A := f(u) = \phi(\alpha)$, $B := f(v) = \phi(\beta)$, $X := f(x)$, $Y := f(y)$, $Z := f(z)$, $P := f(p)$. From $\alpha x = v$ and $y\beta = u$ we get

$$AX = B \;\Rightarrow\; \mathrm{Im}(B) \subseteq \mathrm{Im}(A), \qquad YB = A \;\Rightarrow\; \ker(B) \subseteq \ker(A).$$

If $\dim N = d < \infty$, rank–nullity gives

$$\dim \mathrm{Im}(A) + \dim \ker(A) = d = \dim \mathrm{Im}(B) + \dim \ker(B),$$

hence

$$(\dim \mathrm{Im}(A) - \dim \mathrm{Im}(B)) + (\dim \ker(A) - \dim \ker(B)) = 0.$$

Both brackets are ≥ 0 by the inclusions, so both are 0, and therefore $\mathrm{Im}(A) = \mathrm{Im}(B)$. Now $z\beta = 0$ gives $ZB = 0$, so $\mathrm{Im}(B) \subseteq \ker(Z)$, hence also $\mathrm{Im}(A) \subseteq \ker(Z)$, i.e. $ZA = 0$. But $z\alpha = p$ gives $ZA = P$, so $P = 0$, i.e. $f(p) = 0$ with $p \neq 0$, contradicting injectivity. □

Injectivity in the universal quotient

Proposition 2 (Injectivity into the universal quotient)**.** *The map $\iota : M \to T_R(M)/(e-1)$ is injective.*

Proof. Let H have basis $\{a_i, b_i\}_{i\geq 0} \cup \{c\}$ and define endomorphisms (unspecified values $= 0$):

$$A(a_i) = b_i, \quad B(a_i) = b_{i+1}, \quad X(a_i) = a_{i+1}, \quad Q(a_i) = b_{i+2},$$
$$Y(b_0) = 0,\; Y(b_i) = b_{i-1}\ (i \geq 1), \qquad W(a_0) = 0,\; W(a_i) = b_{i-1}\ (i \geq 1),$$
$$Z(b_0) = c,\; Z(b_i) = 0\ (i \geq 1), \qquad P(a_0) = c,\; P(a_i) = 0\ (i \geq 1), \qquad I := \mathrm{id}_H\,.$$

Then $A^2 = B^2 = AB = BA = 0$, so $\alpha \mapsto A$, $\beta \mapsto B$ defines a left R-module structure on H. Define $\rho : M \to \mathrm{End}_k(H)$ by

$$\rho(e) = I,\; \rho(u) = A,\; \rho(v) = B,\; \rho(x) = X,\; \rho(y) = Y,\; \rho(z) = Z,\; \rho(p) = P,\; \rho(q) = Q,\; \rho(w) = W.$$

One checks directly that the defining bimodule relations above hold:

$$AX = B, \quad BX = Q, \quad YB = A, \quad YA = W, \quad ZA = P, \quad ZB = 0,$$

and all other α, β-actions vanish; hence ρ is an R-bimodule map with $\rho(e) = I$. By the universal property, ρ extends to an algebra map $T_R(M) \to \mathrm{End}_k(H)$ killing $(e-1)$, thus factoring through

$$\bar{\rho} : T_R(M)/(e-1) \to \mathrm{End}_k(H).$$

Finally ρ is injective: if

$$c_I I + c_A A + c_B B + c_X X + c_Y Y + c_Z Z + c_W W + c_P P + c_Q Q = 0,$$

evaluating on c gives $c_I = 0$; on b_1 gives $c_Y = 0$; on b_0 gives $c_Z = 0$; on a_0 gives $c_A = c_B = c_X = c_P = c_Q = 0$; on a_1 gives $c_W = 0$. Thus $\rho|_M$ is injective, and since $\bar{\rho} \circ \iota = \rho$, the map ι is injective. □